A CRM Application for the Chai Shop & CO

USER STORY:

In the CRM Application for the Chai Shop & CO, as a shop owner, manager, or system administrator, I want to use a centralized Salesforce-based platform to record and manage customer, product, order, supplier, and sales information so that all shop data is stored securely and systematically. I want to enter and update customer details, product records, inventory, and customer orders easily through a user-friendly interface with proper validation rules. I also want the system to generate real-time reports and interactive dashboards so that I can analyze sales, monitor inventory, and make informed decisions. Additionally, I want role-based access and secure data storage to protect sensitive information while enabling smooth collaboration.

1. INTRODUCTION

1.1 Project Overview

The CRM Application for the Chai Shop & CO is a Salesforce-based application designed to digitize and streamline chai shop operations. It manages data related to customers, products, orders, inventory, suppliers, and sales while providing real-time insights through reports and dashboards.

1.2 Purpose

The purpose of the CRM application is to replace traditional manual records with an automated cloud-based solution. It helps reduce data loss, provides inventory alerts, supports sales tracking, and offers mobile accessibility while maintaining secure data through Salesforce.

1.3 Existing System & Proposed System

Existing System: Manual notebooks and Excel sheets - limited security, data loss, manual calculations, and no real-time dashboards.
Proposed System: Salesforce cloud platform with custom objects, automation, validation rules, reports, and dashboards. It is low-code, secure, scalable, and mobile-friendly.

2. IDEATION PHASE

2.1 Problem Statement

Chai shop owners may face issues such as losing customer and sales records, difficulty tracking exact profit, manual calculation errors, lack of centralized data, and no low-stock alerts for tea, milk, sugar, snacks, and packaging materials.

2.2 Empathy Map Canvas

SAYS: I lose customer and sales records; I need to know my daily profit.
THINKS: I need a simple and affordable CRM application.
DOES: Writes orders in notebooks, calls suppliers, and calculates sales manually.
FEELS: Frustrated by data loss and worried about stock and expenses.
PAIN: Data loss, calculation mistakes, no central data, and no alerts.
GAIN: Centralized data, automatic calculations, dashboards, and email alerts.

2.3 Brainstorming

1. Manual chai shop record keeping
2. Excel-based management
3. Web-based shop CRM
4. Mobile-based shop application
5. Salesforce-based CRM Application for the Chai Shop & CO

Idea Prioritization: Salesforce was selected because it provides a low-code platform, security, cloud access, mobile support, real-time reports, and a Developer Edition for learning and prototyping.

3. REQUIREMENT ANALYSIS

3.1 Customer Journey Map

Awareness: Shop owner realizes the need for a digital CRM.
Exploration: Searches for solutions and explores Salesforce.
Onboarding: Creates customer and product records.
Usage: Adds products, inventory, suppliers, and orders.
Monitoring: Views dashboard, checks sales and low stock.
Sales: Records orders and tracks total amount and profit.
Support: Receives alerts and monitors activities through the dashboard.

3.2 Solution Requirements

Functional Requirements:
- Create Customer, Product, Inventory, Supplier, Order, and Sales objects.
- Establish Master-Detail and Lookup relationships.
- Validation Rules: Quantity > 0; Order Date should not be a future date.
- Record-Triggered Flow for order and stock automation.
- Reports: Daily Sales, Low Inventory, Top Products, Monthly Sales.
- Dashboard: Total Customers, Sales by Product, Low Stock, Monthly Sales Trend.

Non-Functional Requirements:
- Desktop and Salesforce Mobile App support.
- Role-based secure access.
- Scalable for growing customer and order records.
- User-friendly Lightning UI.
- Fast report and dashboard loading.

3.3 Data Flow Diagram

Level 0 DFD: Customer/Shop Staff (External Entity) → CRM Process (Central System) → Salesforce Database (Data Store) → Reports/Dashboard (Output).

Level 1 DFD:
1.0 Customer Management → Customer Object → DB
2.0 Product Management → Product Object → DB
3.0 Order Management → Order Object → Flow → Sales Object
4.0 Inventory Management → Inventory Object → Low Stock Check
5.0 Supplier & Sales Management → Supplier/Sales Objects → Reports

3.4 Technology Stack

Frontend: Salesforce Lightning Experience, Lightning App Builder.
Backend: Salesforce Platform, Custom Objects, Validation Rules, Flows.
Database: Salesforce Cloud Database.
Tools: Developer Org, Setup Menu, App Manager, Object Manager, Flow Builder, Report Builder.

4. PROJECT DESIGN

4.1 Problem-Solution Fit

Problem: Data loss, no clear sales/profit tracking, manual errors, and stock management difficulties.
Customer Segment: Small and medium chai shops, shop owners, and managers.
Existing Alternatives: Manual books and Excel sheets.
Solution: Salesforce CRM solves these pain points with automation, security, centralized records, and dashboards.

4.2 Proposed Solution

We propose a Lightning Application named **"Chai Shop & CO CRM"** containing 6 custom objects.

Object Structure:
Customer (Parent) - One to Many - Order
Product (Parent) - One to Many - Order
Supplier (Parent) - One to Many - Inventory
Order (Parent) - One to Many - Sales/Order Items
Product - One to Many - Inventory

Fields: Product Category (Chai, Coffee, Snacks, Food, Other), Order Status (New, Preparing, Completed, Cancelled).
Formula Field: Total_Amount = Quantity × Unit_Price.
Automation: When an order is completed, the Flow updates stock, creates the sales record, and sends an email notification where required.

4.3 Solution Architecture

Presentation Tier: User Interface (Lightning Tabs, App, Mobile App)
Business Logic Tier: Validation Rules, Record-Triggered Flows, Approval Process
Data Tier: Custom Objects (Customer, Product, Inventory, Supplier, Order, Sales)

Architecture Flow: End User (Browser/Mobile) → Salesforce Cloud Platform → Lightning App → Custom Objects → Salesforce Database → Reports & Dashboards → End User.

5. PROJECT PLANNING & SCHEDULING

Sprint 1 (Week 1): Requirement Gathering, Empathy Map, Brainstorming, User Story Documentation.
Sprint 2 (Week 2): Create Salesforce Developer Org, Data Modeling, Custom Objects and Fields, Relationships.
Sprint 3 (Week 3): Create Custom Tabs, Lightning App, Validation Rules, Page Layouts.
Sprint 4 (Week 4): Build Flows, Reports and Dashboards, Testing, Documentation and GitHub Upload.

6. PROJECT DEVELOPMENT PHASE

6.1 Development Steps

Step 1: Created Salesforce Developer Org.
Step 2: Created Custom Objects:

Customer: Name, Phone, Email, Address, Customer_Type.
Product: Product_Name, Category, Unit_Price, Available_Quantity, Status.
Inventory: Item_Name, Category, Quantity, Unit_Price, Supplier, Product.
Supplier: Supplier_Name, Phone, Email, Address.
Order: Customer, Product, Quantity, Unit_Price, Total_Amount, Order_Date, Status.
Sales: Order, Product, Quantity_Sold, Rate, Total_Amount, Buyer/Customer, Sale_Date.

Step 3: Created Custom Tabs for all objects via Setup → Tabs.
Step 4: Created Lightning App: Setup → App Manager → New Lightning App → Name: Chai Shop & CO CRM → Add all Tabs → Assign to appropriate profile.
Step 5: Created Validation Rules for positive quantity and valid order dates.
Step 6: Created Record-Triggered Flow for completed orders, stock updates, sales record creation, and email alerts.

Step 7: Created Reports: Daily Sales, Low Inventory, Top Products, Monthly Sales.
Step 8: Created Dashboard: Total Customers, Sales by Product, Low Stock Inventory, Monthly Sales Trend.

7. FUNCTIONAL AND PERFORMANCE TESTING

7.1 Functional Testing

TC01: Create Customer record with valid data - Expected: Saved - Result: Passed
TC02: Create Product record - Expected: Saved - Result: Passed
TC03: Enter invalid quantity - Expected: Validation error - Result: Passed
TC04: Complete an order - Expected: Flow updates stock and sales - Result: Passed
TC05: View Dashboard - Expected: Charts display - Result: Passed
TC06: Mobile App access - Expected: App opens in Salesforce mobile - Result: Passed

7.2 Performance Testing

Tested with sample customer, product, inventory, and order records. Page, report, and dashboard performance were checked to ensure smooth operation.

8. ADVANTAGES & DISADVANTAGES

Advantages:
1. Centralized cloud-based CRM data management.
2. Low-code platform with reduced development effort.
3. Real-time reports and dashboards.
4. Mobile accessibility via Salesforce Mobile App.
5. Automation reduces manual work and errors.
6. Secure access using Salesforce security features.
7. Scalable for growing shop operations.
8. Developer Edition is useful for learning and prototyping.

Disadvantages:
1. Requires internet connectivity.
2. Salesforce Enterprise licenses can be costly.
3. Developer Edition has storage and feature limits.
4. Users need basic digital literacy.
5. Customization can be more limited than a full-code application for some requirements.

9. CONCLUSION

The CRM Application for the Chai Shop & CO is designed as a Salesforce-based solution to digitize chai shop operations, centralize customer and sales information, manage products and inventory, and provide real-time insights through reports and dashboards. The project demonstrates practical use of Salesforce Administration, Data Modeling, Automation using Flows, Reports, and Dashboards. The application can be extended with additional features such as loyalty management, online ordering, supplier purchase tracking, and customer feedback.